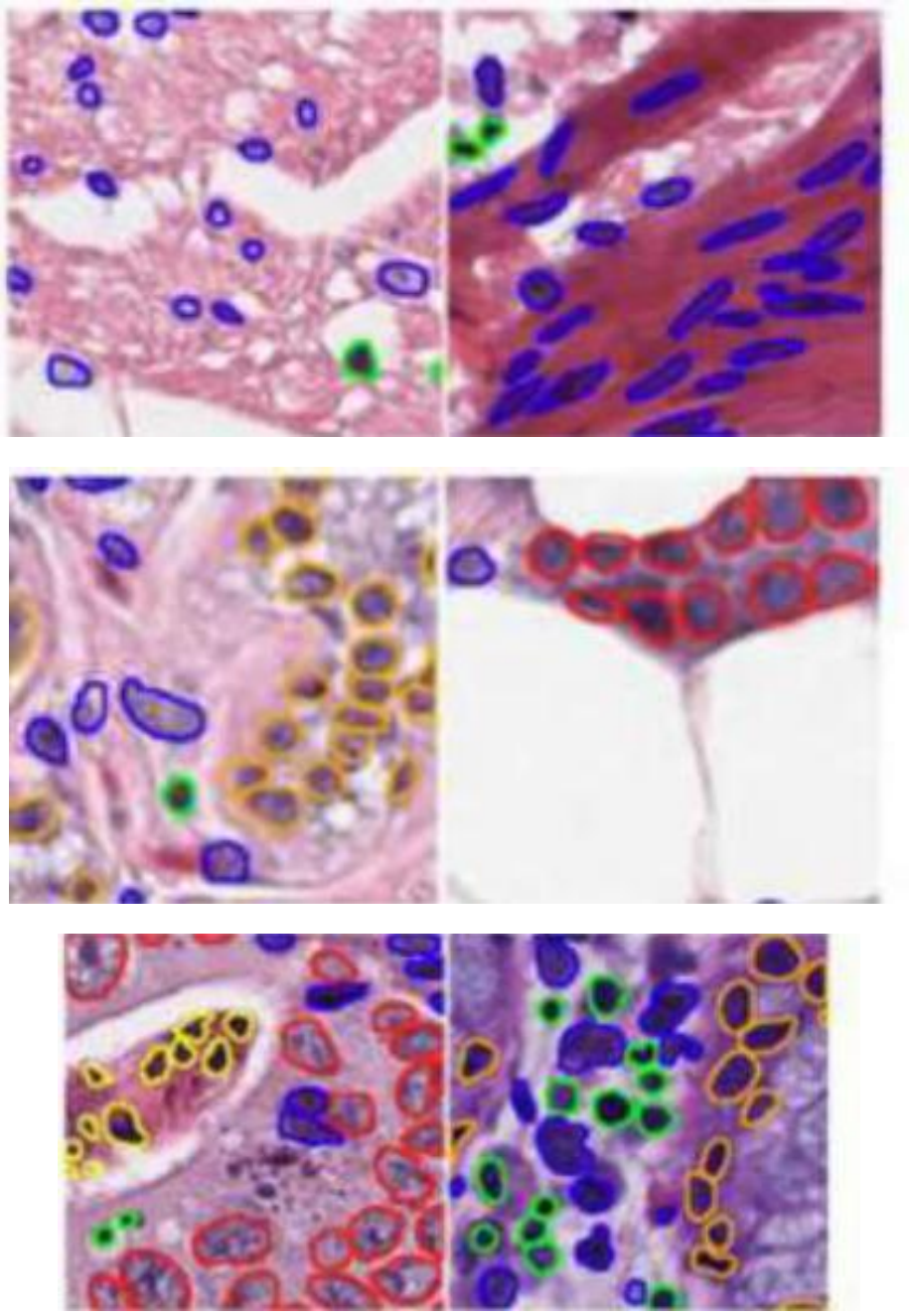
BIOMEDICAL DATA DESIGN I

Nuclei instance segmentation across diverse tissues

HoVer-Net, CellViT-UNI, and PM-Net

Weekly report • October 15, 2026

Xinhao Jin



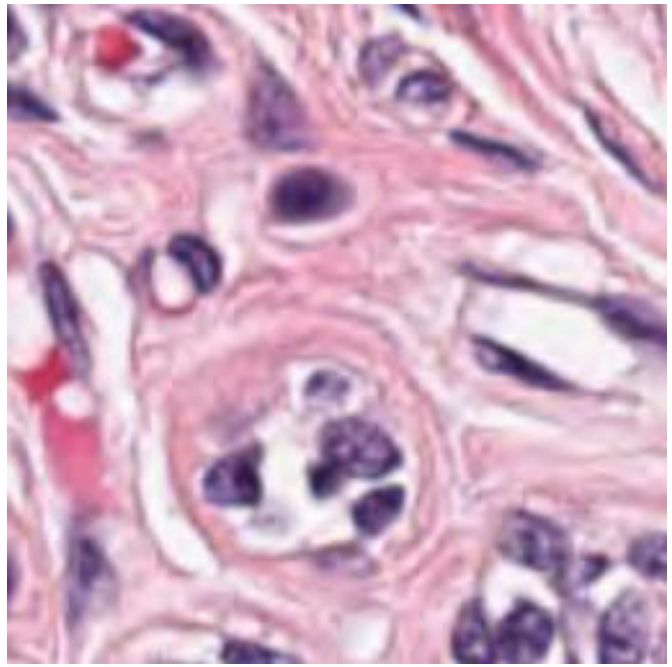
Three parts of today’s update

1	**Model structures**	HoVer-Net, UNI, and the CellViT decoder
2	**Completed experiments**	What we tried, what happened, and why
3	**PM-Net proposal**	Direct instance masks and a controlled test

THE TASK

A nucleus needs its own mask and class label

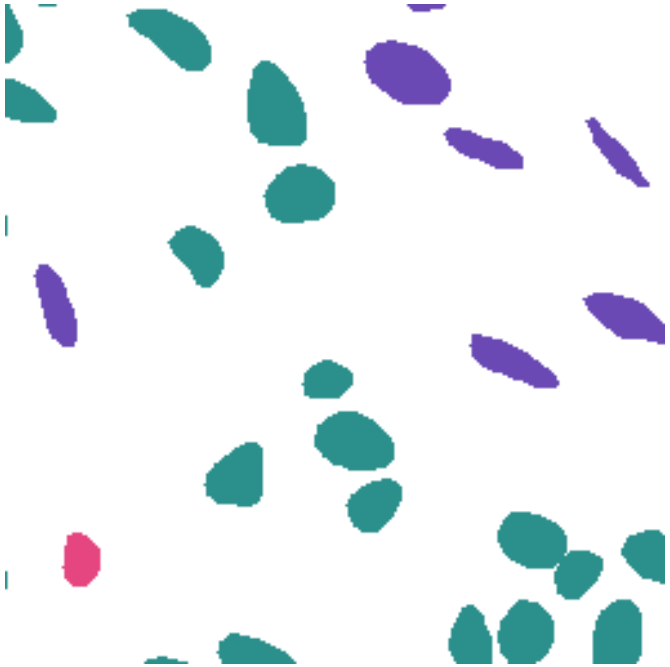
H&E image



Individual instances

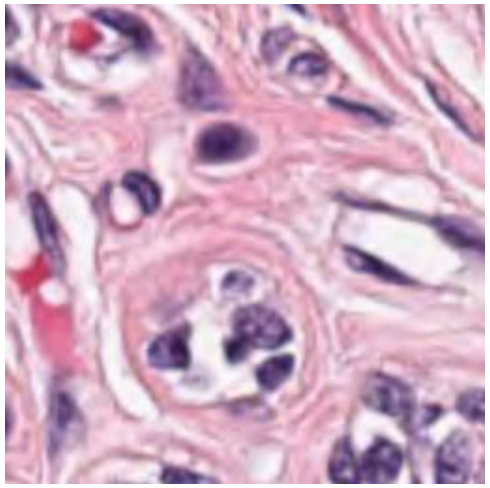


Five nucleus classes



Touching nuclei must remain separate objects.

The encoder finds features; the decoder restores detail



Encoder
Image features

Texture, shape,
and tissue context

Decoder
Pixel predictions

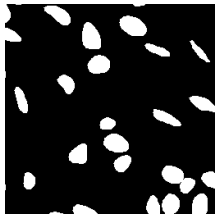
Where the nucleus is
and how its boundary looks

Good features still need a reliable path back to image coordinates.

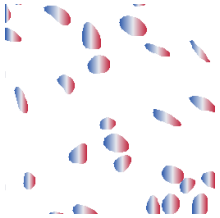
One shared encoder feeds three decoder branches

CNN encoder ResNet-50

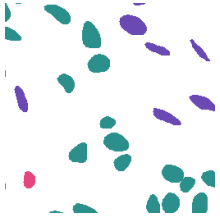
NP: nucleus pixels



HV: relative position



TP: nucleus type



Separate jobs,
shared image features

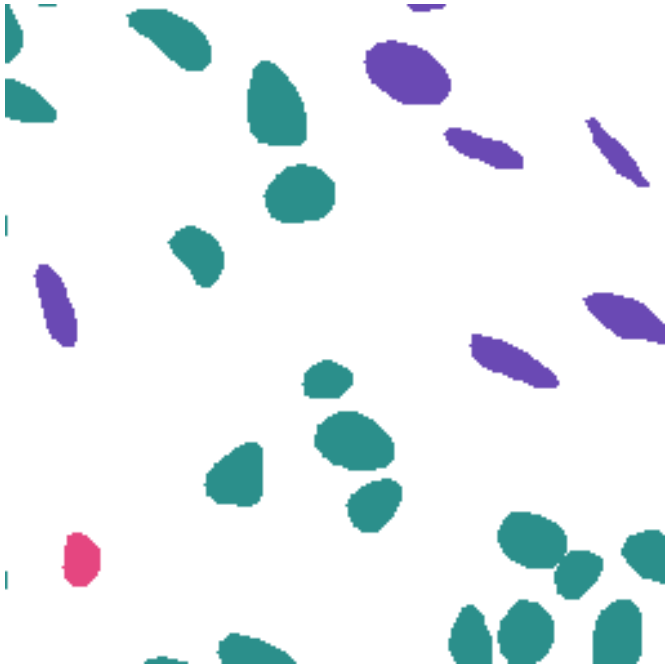
NP and TP are pixel classification. HV predicts two continuous values.

Foreground and type maps answer different questions

NP



TP



Is this pixel inside a nucleus?

Which nucleus type does it resemble?

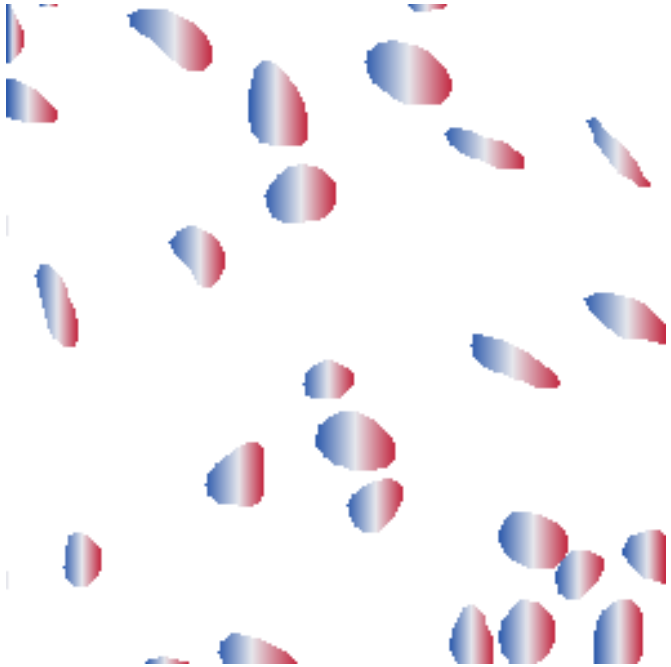
A foreground region may still contain more than one nucleus.

Each nucleus has its own horizontal and vertical coordinates

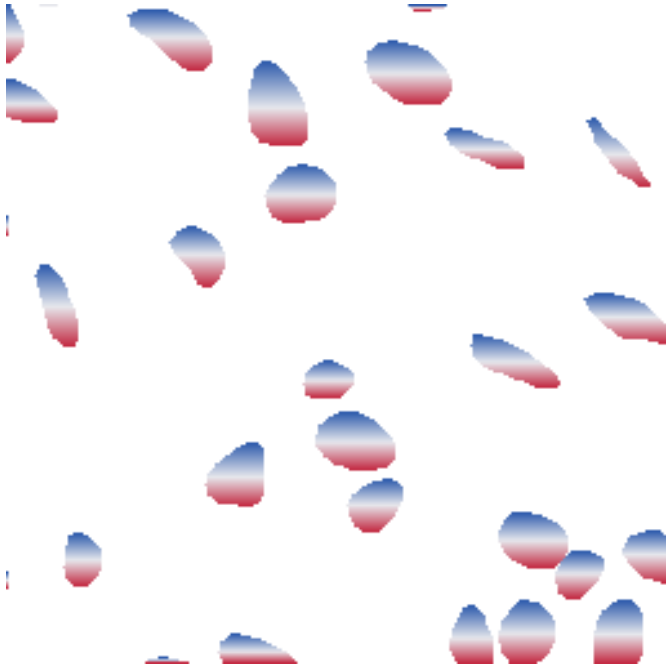
Instance IDs



Horizontal offset



Vertical offset

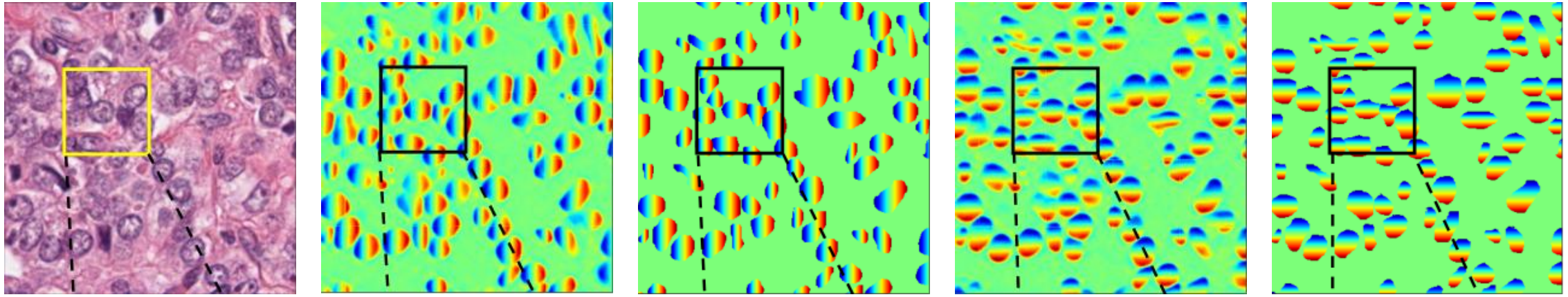


Blue = negative offset. Red = positive offset. White background is outside nuclei.

PAPER EXAMPLE

Real HV predictions preserve useful separation cues

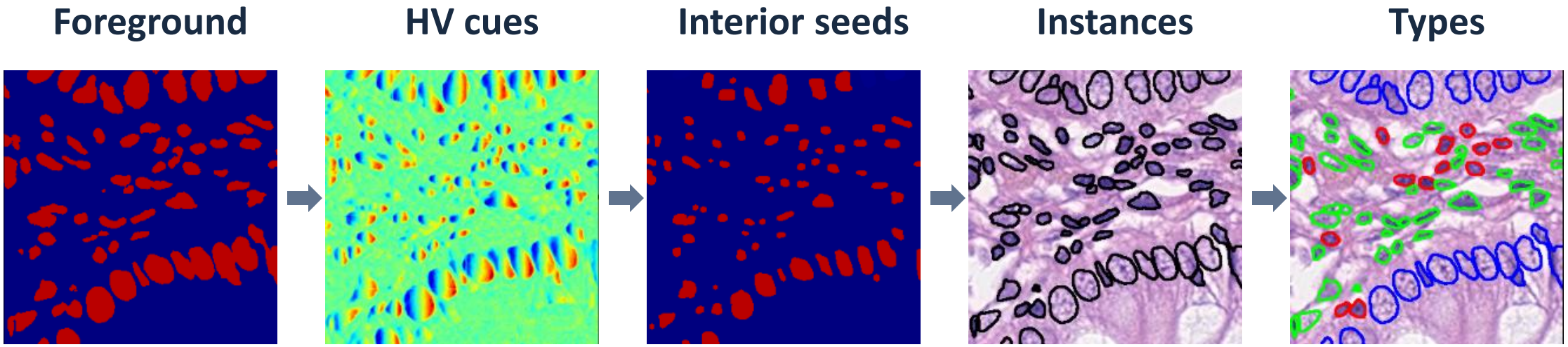
Tissue image | H prediction | H target | V prediction | V target



Prediction smoother and imperfect

A sharp coordinate change can reveal where two touching nuclei separate.

Watershed turns pixel maps into separate objects



Original prediction and decoding panels from Graham et al. (2019), Fig. 1.

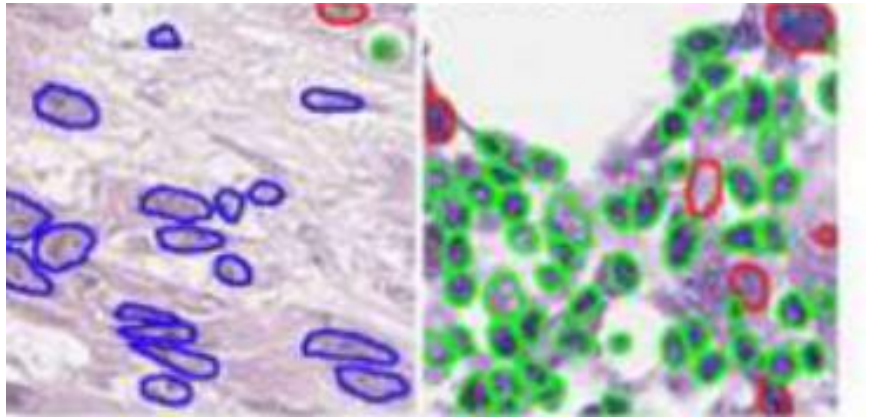
Missing seeds can merge nuclei. Extra seeds can split one nucleus.

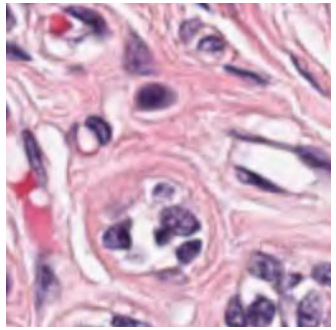
Three losses teach foreground, geometry, and type

Branch	Target	Training signal
NP	Nucleus / background	Classification + Dice
HV	Horizontal / vertical offsets	Value + gradient error
TP	Background + five classes	Classification + Dice

When a tiny object loses HV targets, its NP and TP supervision still remains.

UNI learns pathology features before our task training





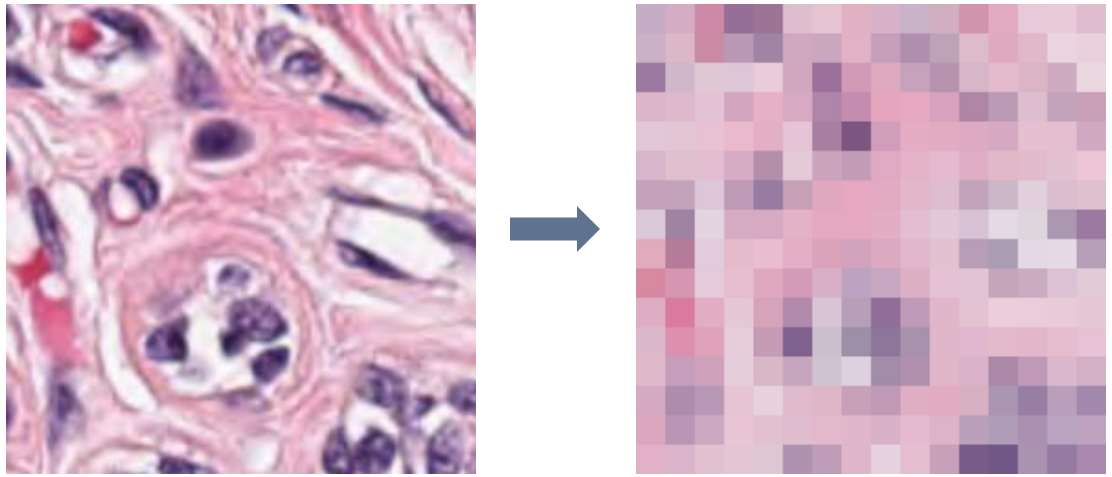
More than 100 million image patches

Over 100,000 H&E whole-slide images

Self-supervised pretraining
across 20 major tissue types

UNI supplies image features. The nucleus decoder still has to learn our task.

A 256-pixel patch becomes a 16 × 16 feature grid



256 × 256 input

16 × 16 patches

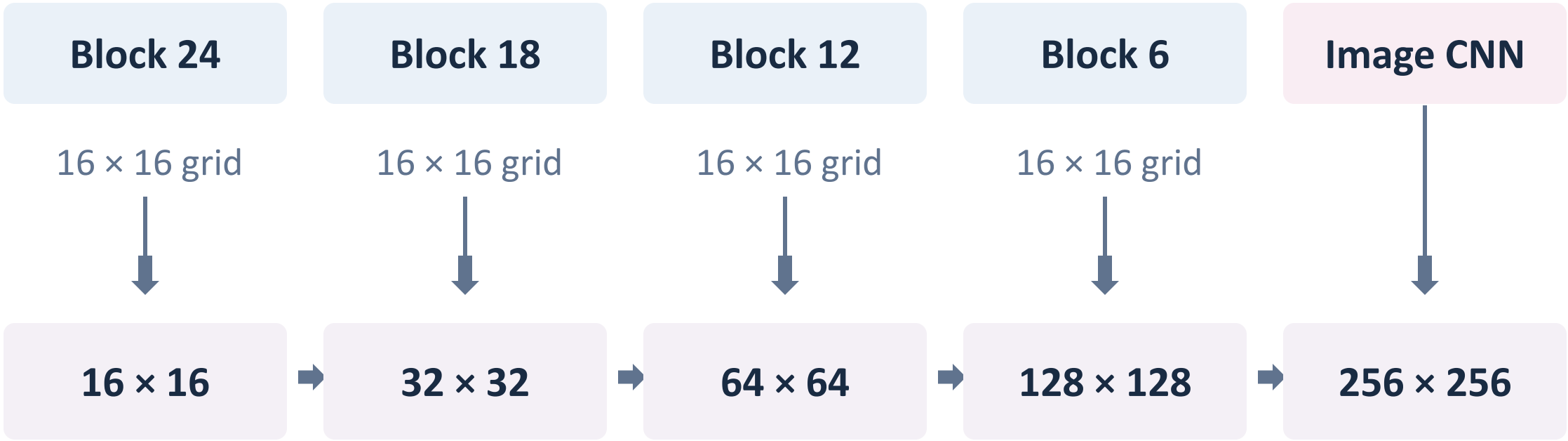
24 ViT blocks

256 spatial tokens
1024 values per token

Grid image illustrates patch sampling, not a measured UNI activation.

We extract features after blocks 6, 12, 18, and 24.

Features from four depths recover fine spatial detail

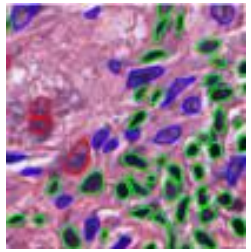


Upsampling restores resolution; skip features supply detail at each stage.

A stronger encoder keeps the same three-output recipe

UNI encoder → **Upsampling + skip fusion** → **NP / HV / TP**

NP / HV / TP → **Watershed + instance typing**



Paper output example

Better features do not remove foreground, separation, or typing errors.

EVALUATION

Detection, mask quality, and type all affect the score

mPQ	**Type-aware quality**	Class-specific PQ, averaged across five classes
bPQ	**Instance quality**	All nuclei treated as one foreground class
Dead PQ	**Rare-class quality**	Focus on dead nuclei
Strict scores	**Absent-class penalties**	Retain false positives when a class is absent

An official Dead gain can coexist with worse strict Dead or bPQ.

BASELINE EVIDENCE

CellViT-UNI improves mPQ, but Dead misses remain

Own training, no TTA	mPQ	bPQ
HoVer-Net	0.4564	0.6613
CellViT-UNI	0.4995	0.6653

38.5%

of Dead nuclei were missed in the original baseline diagnosis

Three-split means above. The miss-rate diagnosis is a separate historical analysis.

EXPERIMENT MAP

Completed studies tested four possible explanations

Question	Studies	Current finding
Better typing or data?	CONCH, T0–T3, synthesis, CF	No stable repair from tested recipes
Different instances or scale?	KongNet, LKCell, P0–P3, EF-P2	Some gains, quality or cost tradeoffs
Thresholds, labels, candidates?	E2a, DSB, E0, E1, M, E3	Narrower explanations, open uncertainty
Crop and target supervision?	E4, four arms, fixed C2	Tested intervention fails its criteria

The project has useful evidence, but no broadly superior replacement method.

Typing headroom did not become a useful retyping method

CONCH: image-text features for existing nuclei

Balanced accuracy 0.695; fused mPQ only 0.4951 to 0.4953.

T0–T3: new classifiers on frozen features

Best ΔmPQ +0.0002; a linear head loses 0.0053 Dead PQ.

GT-assisted typing oracle

mPQ 0.4995 to 0.6632, using labels unavailable at inference.

A classification ceiling is not an achievable method gain.

The tested generator did not repair the detection deficit

Failure-driven augmentation

Real Dead copy-paste: ΔDead PQ −0.031. Diffusion: mPQ 0.4900.

Counterfactual rendering

Keep labels, change appearance. Overall cross-domain rank correlation: −0.89.

Interpretation

Binary-mask conditioning limits class control; detection gaps were not reproduced.

The current renderer is a limited typing stress test, not a transfer proxy.

More detections can come with more merges or false positives

Model or study	Result	What it suggests
HoVer-NeXt-T	mPQ 0.4579	Released recipe under our evaluator
LKCell-L	mPQ 0.4923	Better binary masks, weaker typing
KongNet	mPQ 0.3460	Touching same-class nuclei often merge
2× scale, P0–P3	Extra Dead candidates	Decode scaling and training scale matter

Released weights have different training / selection histories. These are not matched architecture ablations.

Filtered scale additions keep a small Dead gain at high cost

Nine matched split / seed pairs	mPQ	bPQ	Dead PQ
Matched baseline	0.49760	0.66530	0.17255
P2 additions	0.49769	0.66443	0.17997
EF-P2 filtered additions	0.49828	0.66529	0.17954

+0.0070 Dead PQ

≈6.1× inference cost

Image-bootstrap intervals span zero. Historical test folds were reused across rounds.

MECHANISM CONTROLS

HV area thresholds and class conflict are incomplete explanations

E2a: scale × HV target support, four arms and two seeds

Lowering the x1 cutoff reproduces only 9.6% of the x2 Dead gain.

DSB: test the need for a separate Dead expert

Mean gradient cosine +0.749; none of 64 valid batches is negative.

Decision

Keep HV cutoff 30. Stop the specialist route before expert training.

The measured class-conflict premise failed under the tested conditions.

Detector agreement is useful evidence, but not ground truth

E1-C / E1-T: agreement across detectors

66–73% candidate support; 72.5% vote for the base model's disputed type.

M: merge inflammatory and connective labels

Official mPQ changes by +0.0519 in a different label space.

E3: learn candidate utility

AUC 0.721 / 0.737, but strict Dead criteria fail. Identity fallback.

Expert review remains unavailable. Five-class method scores are unchanged by M.

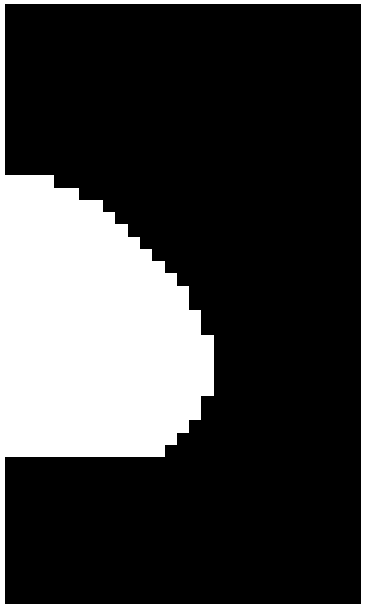
CROP SENSITIVITY

E4 showed a vulnerability, then motivated an intervention

Visible nucleus



Cropped mask



+0.1738

Area-standardized edge–interior Dead miss gap

Crop vulnerability does not prove that zero HV targets cause misses.

Four arms separate stronger cropping from selective HV targets

Arm	Crop probability	HV target rule
A0: retrained baseline	0.1	Minimum area 30
A1: crop-only control	0.5	Minimum area 30
A2: selective targets	0.5	Minimum 1 for new small crop fragments
A3: all small objects	0.5	Minimum area 1 for all objects

A2 – A1 isolates the added value of selective HV supervision.

SUPERVISION RESULTS

The original-image quality criteria fail

Mean change, two seeds	mPQ	bPQ	Dead PQ	Strict Dead
A1 − A0	−0.0053	−0.0061	+0.0376	+0.0088
A2 − A0	−0.0063	−0.0054	+0.0246	−0.0009
A2 − A1	−0.0010	+0.0007	−0.0131	−0.0098

Selective HV adds no Dead benefit over crop-only training.

Validation only. A2 exceeds the 0.002 mPQ / bPQ loss tolerance in both seeds.

The crop-mechanism criterion also fails

−0.0086

Mean A1 − A2 miss-rate improvement

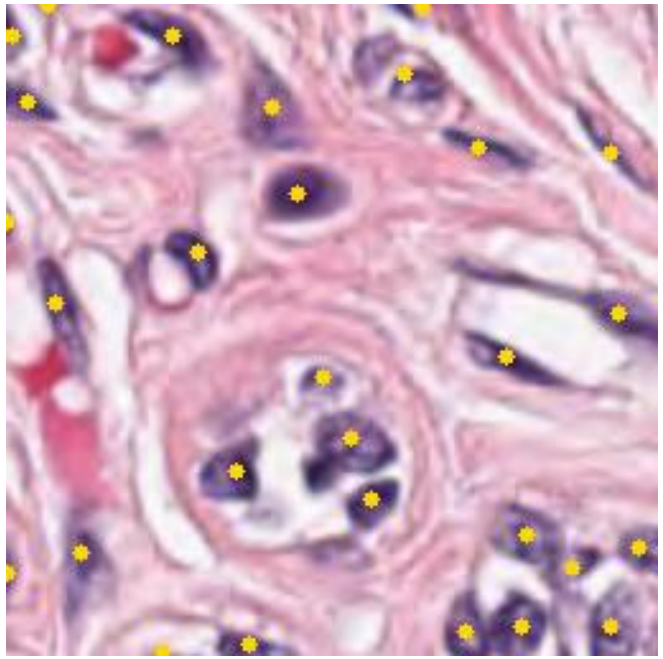
95% interval: [−0.0409, +0.0177]

116 eligible Dead objects in 38 images; inference covered 2,523 fixed windows.

Stop this dev recipe. No extra seeds or three-split confirmation.

Direct masks test a different instance representation

Candidate locations



One query
One proposed nucleus

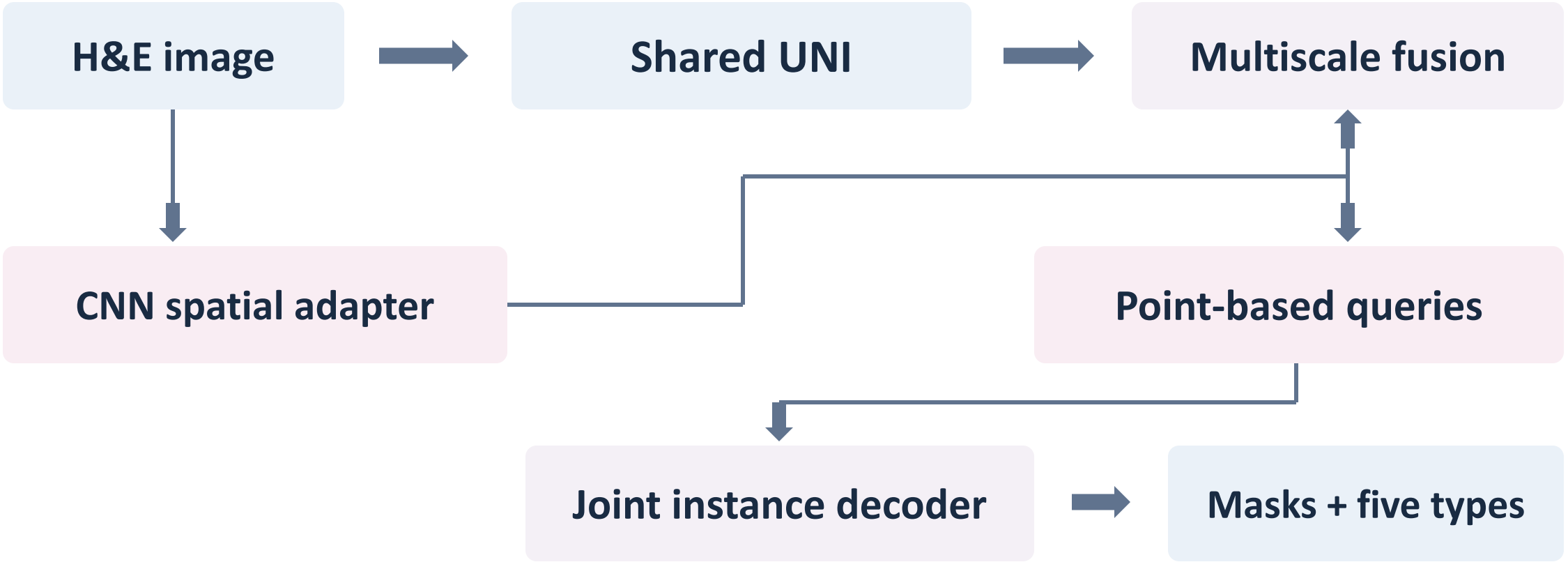
Instance masks



Hypothesis: improve detection and separation without a second full-scale model.

PM-NET ARCHITECTURE

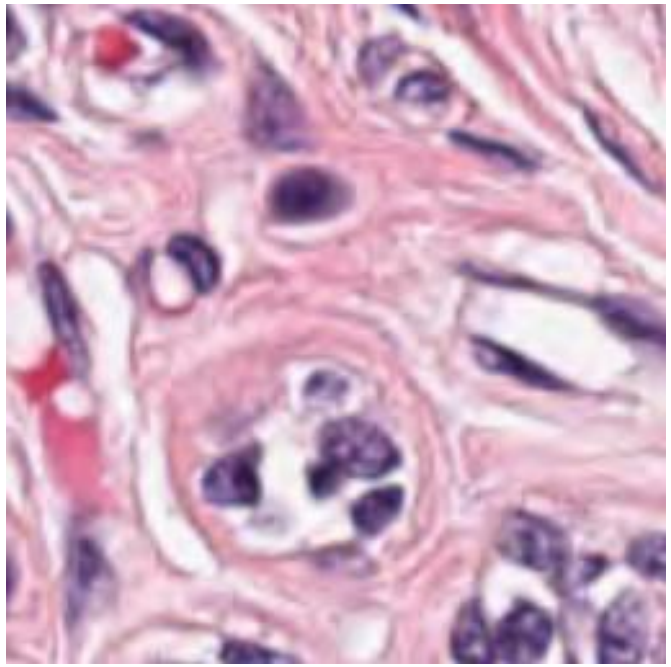
One image encoding supports all nucleus queries



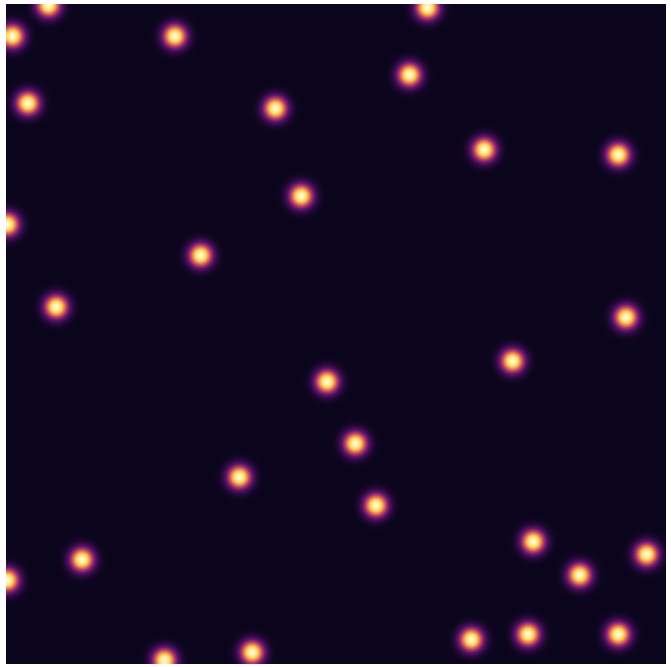
Proposed model. No HV map or watershed in the instance decoder.

A point target must stay inside the visible nucleus

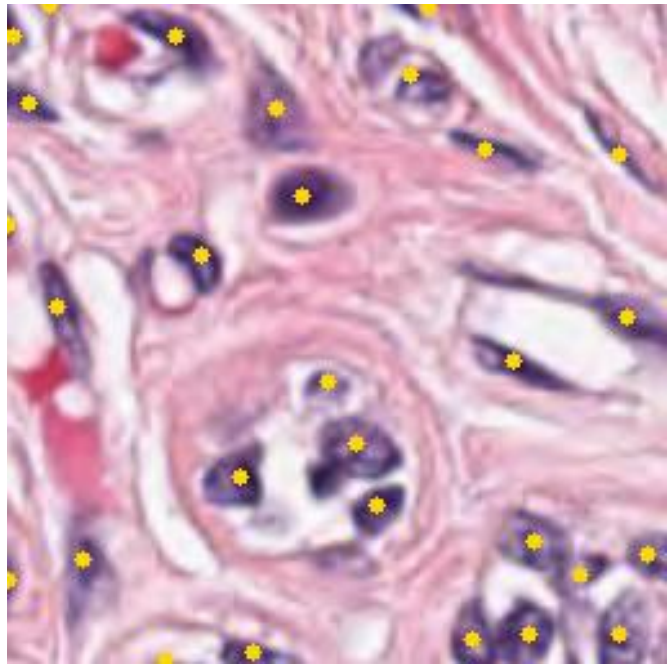
Image



Illustrative point target



Inside-mask points



Direct heatmap supervision trains the point head despite discrete point selection.

One-to-one matching links queries to labeled nuclei



Match using location, class, and mask agreement. Train unmatched queries as no object.

A detected point still needs a mask with sufficient overlap.

Point initialization and fusion must earn their added cost

Comparison	Question
Basic queries vs point-initialized queries	Does point initialization help?
Plain vs gated multiscale fusion	Does the fusion rule help?
Parameter-matched ordinary adapter	Is the gain only extra capacity?

276 objects in one training patch

256 queries are insufficient. 512 covers original labels, but cost is untested.

Also handle 23 rounded centers outside masks and 116 empty-label training images.

EVALUATION PLAN

Detection gains must preserve quality and inference cost

Endpoint	Proposed development requirement
Interior Dead recall	Mean gain ≥ 0.03; paired lower 95% bound > 0
Dead PQ	Mean gain ≥ 0.005; both seeds improve
mPQ / bPQ	Per-seed loss no more than 0.002
Strict scores and instance errors	No strict decline; mean merge/split rates do not rise
End-to-end latency	At most 2× the single-scale baseline

Matched UNI initialization, schedule, seeds, final checkpoints, and evaluator.

NEXT STEPS

A staged plan leaves a useful result even if PM-Net fails

1	**Minimum**	Baselines, error analysis, and completed supervision results
2	**Target**	A tested PM-Net prototype with two-seed ablations
3	**Stretch**	Three-split and external confirmation if dev criteria pass

The next study tests the instance representation, with clear quality and cost limits.